

Nassau Candy Distributor

Final Business Analytics & Profitability Report

Submission-ready analysis covering data quality, EDA, division performance, product profitability, Pareto concentration, cost & margin diagnostics, and management actions.

1. Executive Summary

Metric	Result
Total Sales	\$141,783.63
Total Cost	\$48,340.83
Gross Profit	\$93,442.80
Units Sold	38,654
Gross Margin	65.91%

- The dataset contains 10,194 records and 15 products. Total sales are \$141,783.63, gross profit is \$93,442.80, and weighted gross margin is 65.91%.
- Chocolate is the dominant division, contributing approximately 92.88% of revenue and 95.06% of gross profit.
- The highest-margin product is Everlasting Gobstopper at 80.00%.
- The key margin-risk product is Kazookles at 7.69%.
- Five products reach/exceed both the 80% cumulative revenue and cumulative profit thresholds.

2. Data Quality

Check	Result
Total records	10194
Duplicate records flagged	620
Duplicate groups	307
Missing/invalid Sales	0
Missing/invalid Cost	0
Missing/invalid Gross Profit	0
Gross Profit mismatch rows	0
Invalid/zero Units	0
Rows with Shipping Days >30	10194

The source contains a retained shipping-date quality flag: unusually long shipping intervals should be validated before operational service-level use.

3. Division Performance

Division	Sales	GP	Margin	Revenue %	Profit %
CHOCOLATE	\$131,692.90	\$88,824.62	67.45%	92.88%	95.06%
Other	\$9,663.25	\$4,333.45	44.84%	6.82%	4.64%
Sugar	\$427.48	\$284.73	66.61%	0.30%	0.30%

Chocolate is the core economic engine. Other shows materially higher cost pressure and lower margin; Sugar has a healthy margin but very small scale.

4. Product Profitability & Pareto

#	Product	Sales	GP	Margin	Risk
1	Wonka Bar -Scrumdiddlyumptious	\$27,874.80	\$19,357.50	69.44%	Low
2	Wonka Bar - Triple Dazzle Caramel	\$28,485.00	\$18,610.20	65.33%	Watch
3	Wonka Bar - Milk Chocolate	\$26,867.75	\$17,443.37	64.92%	Watch
4	Wonka Bar - Nutty Crunch Surprise	\$23,574.95	\$16,819.95	71.35%	Low
5	Wonka Bar - Fudge Mallows	\$24,890.40	\$16,593.60	66.67%	Low
6	Lickable Wallpaper	\$7,860.00	\$3,930.00	50.00%	Watch
7	Wonka Gum	\$597.50	\$310.70	52.00%	Watch
8	Everlasting Gobstopper	\$130.00	\$104.00	80.00%	Low
9	Kazookles	\$1,205.75	\$92.75	7.69%	High
10	Hair Toffee	\$76.50	\$59.50	77.78%	Low

The top 10 products generate 99.87% of gross profit, showing substantial concentration.

5. Cost & Margin Diagnostics

Company gross-margin benchmark: 65.91%. Kazookles is the most significant margin exception at 7.69%. Its economics should be reviewed across selling price, discounting, purchase cost, and unit profitability.

6. Final Management Actions

- Protect availability and pricing discipline for the core products driving most revenue and gross profit.
- Review Kazookles first for price, discount, supplier cost and unit economics; establish a corrective target margin.
- Investigate procurement and cost pressure in the Other division.
- Use the Pareto product set to prioritize inventory, working capital and management attention.
- Validate shipping-date fields before operational reporting.

Conclusion

The project now provides a complete analytics workflow from validated source data through executive insights and actionable profitability diagnostics.